

DAY 4: NETWORK CABLE TYPES AND TOOLS

GitHub Assignment • Ready for Submission

Theory	Types of cables (Ethernet, Fiber Optics) and tools (RJ45 Crimper)
Practical	Create a straight-through Ethernet cable
Status	Completed / Ready for Submission

Objective

To understand common network cable types, identify their uses, learn the T568B wiring standard, and understand how to terminate a CAT6 cable using an RJ45 crimper.

Task 1: Ethernet vs Fiber Optic

Difference	Ethernet Cable	Fiber Optic Cable
Transmission	Electrical signals	Light signals
Distance	Short/medium distances	Long distances
Speed & interference	Good speed; may be affected by EMI	Very high speed; immune to EMI
Example	Home/office router, switch and PCs	Data center or campus backbone

Task 2: Straight-Through Ethernet Cable – T568B

A straight-through cable uses the same T568B wiring standard on both ends.

Pin	T568B Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Diagram: End A T568B → End B T568B (same pin order)

END A	END B
1 White/Orange	1 White/Orange
2 Orange	2 Orange
3 White/Green	3 White/Green
4 Blue	4 Blue
5 White/Blue	5 White/Blue
6 Green	6 Green
7 White/Brown	7 White/Brown
8 Brown	8 Brown

TASK 3: RJ45 CRIMPER – CAT6 CABLE

Step-by-step guide

1. Cut the CAT6 cable to the required length.
2. Remove a small section of the outer jacket carefully.
3. Untwist and separate the eight wires.
4. Arrange the wires in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5. Straighten and trim the wires evenly.
6. Insert all eight wires fully into the RJ45 connector.
7. Check the wire order and make sure the cable jacket enters the connector.
8. Place the connector in the RJ45 crimper and squeeze firmly.
9. Use the same T568B order on the other end for a straight-through cable.
10. Test the finished cable with a LAN cable tester.

INSERT RJ45 CRIMPING / CABLE MAKING PHOTO HERE

Task 4: Gaming Café – 10 PCs

I would choose Ethernet CAT6 cables because they are affordable, reliable, and suitable for short distances between the PCs and network switch. Ethernet also provides low-latency connections for LAN gaming. I would use an RJ45 crimper to make and terminate the cables. Fiber optics is better for long-distance backbone links but costs more and needs specialized tools.

Topic Repository URL

GitHub: <https://github.com/your-repo>

Submission Status: **Ready for GitHub Upload**